
Problem 1: Patient Registry

Background: You are building a simple registry of three patients enrolled in a diabetes study: "Rahim", "Karim", "Sultana".

Task: Create a character vector with their names and print the second patient.

Problem 2: Blood Pressure Data

Background: Systolic blood pressure (SBP) values were recorded for four hypertensive patients: 145, 160, 138, 152.

Task: Store these in a numeric vector and extract the **3rd and 4th patients' SBP**.

Problem 3: Gene Expression Counts

Background: Expression counts for the gene *TP53* were measured in four samples: 500, 800, 450, 600.

Task: Create a vector and calculate the mean expression.

Problem 4: Small Cohort Data Frame

Background: A nutrition study enrolled three patients with the following ages and BMI values:

Name Age BMI

Ali 35 24.5

Sonia 29 28.2

Kabir 41 31.0

Task: Create a data frame from this dataset.

Problem 5: Add BMI Classification

Background: For clinical interpretation, classify BMI into:

- "Normal" if < 25
- "Overweight" if $25\text{--}29.9$
- "Obese" if ≥ 30

Task: Add a new column "BMI_Class" to the data frame.

Problem 6: Subset Patients by Age

Background: Researchers want only participants older than 30 years.

Task: Use conditional indexing to filter the data frame.

Problem 7: Subset Patients by BMI

Background: The team wants to see patients classified as "Obese".

Task: Filter the data frame to select only those rows.

Problem 8: Create a Gene Expression Matrix

Background: You measured expression of three genes (*BRCA1*, *EGFR*, *MYC*) across three patients.

Gene P1 P2 P3

BRCA1 120 150 130

EGFR 300 350 400

Gene P1 P2 P3
MYC 800 900 950

Task: Create this as a matrix with proper row and column names.

Problem 9: Extract Specific Expression Value

Background: You want *MYC*'s expression for **Patient 2**.

Task: Extract this value from your matrix.

Problem 10: Create a Multi-Object List

Background: You want to store different analysis objects together:

1. A vector of patient ages: 35, 29, 41
2. The gene expression matrix from Problem 8
3. The nutrition data frame from Problem 4

Task: Create a list with these three objects.

Problem 11: Extract from a List

Background: You want to analyze *EGFR* expression across all patients.

Task: Extract the second row of the matrix from the list.

Problem 12: Conditional Statement for Glucose Check

Background: Fasting blood glucose (FBG) was measured for a patient and stored as *x*.

- If $FBG < 100 \rightarrow$ "Normal"
- If $FBG 100-125 \rightarrow$ "Prediabetes"
- If $FBG \geq 126 \rightarrow$ "Diabetes"

Task: Write an if-else statement to classify the patient based on *x*.

Medium to Hard

Problem 13: Identify High-Risk Patients

Background: A cardiovascular study defines “high risk” as patients who are both **older than 40 years** AND have **BMI ≥ 30** .

Name Age BMI

Tania 42 31.5

Mahir 37 28.0

Jui 45 29.5

Imran 50 32.0

Task:

1. Create this as a data frame.
 2. Use conditional indexing to extract patients classified as “high risk.”
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Problem 14: Integrate Different Biomedical Data in a List

Background: In an oncology project, you need to store:

- A patient demographic data frame (Name, Age, Sex).
- A matrix of tumor marker expression values (CA125, CEA, PSA across 3 patients).
- A vector of survival times (months).

Task:

1. Create a list containing all three objects.
2. Extract:
 - The CEA expression for the 2nd patient.
 - The survival time for the 3rd patient.

Problem 15: Automated Classification of Anemia Status

Background: Hemoglobin (Hb) values were measured for 6 patients: 11.2, 13.5, 9.8, 14.1, 12.0, 8.7.

WHO guidelines:

- $Hb < 12 \rightarrow \text{"Anemia"}$
- $Hb \geq 12 \rightarrow \text{"Normal"}$

Task:

1. Store the Hb values in a vector.
2. Use `ifelse()` to create a classification vector ("Anemia" or "Normal").
3. Add this classification as a new column in a data frame of patients.